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MILITARY SPECIFICATION

PLASTIC MATERIAL, CELLULAR POLYURETHANE, FOAM-IN-PLACE, RIGID (3 POUNDS PER CUBIC FOOT DENSITY)

This specification is approved for use by the Air Force Materials Laboratory, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a rigid closed cell polyurethane foam to be used for the foam-in-place helmet liner process. It also covers the materials required and the preparation of the formulation used in the process.

1.2 Classification. The formulation of materials shall be classified as follows:

Component I - Polyol
Component II - Isocyanate
Component III - Catalyst

1.2.1 One bottle of polyol, one bottle of isocyanate and an appropriate amount of catalyst is sufficient for preparing one foam liner.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Aeronautical Systems Division, ENESS, Wright-Patterson Air Force Base, Ohio 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8030

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SPECIFICATIONS

FEDERAL

PPP-G-460	Glass Containers, One Gallon Capacity and Smaller For Other Than Medical Products, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

T.O 14P3-1-121	Custom Fitting Procedures - Flying Helmet Assembly, Types HGU-2A/P and HGU-26/P

(Application for copies should be addressed to ASD/AEL, Wright-Patterson Air Force Base, Ohio 45433.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Committee

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, ILL 60606.)

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3. REQUIREMENTS

3.1 First article. The cellular polyurethane foam material furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein.

3.2. Materials. The components used in the formulation of polyurethane foam shall conform to the requirements of this specification.

3.3 Components.

3.3.1 Component composition. The composition of components shall be as specified in Table I.

Table I. Composition. 1/

Composition I (Polyol)		Composition II (Isocyanate)	
Pluracol TP-440	160 grams	Mondur MR	190 grams
Stearic acid	0.8 grams	DuPont Freon F-11	21 grams
Dow Corning DC-113	2.4 grams		
DuPont Freon F-11	48 grams		
Component III (Catalyst) Dibutyl Tin Diacetate			

1/ The above components I and II composition will make approximately 2-1/2 bottles (3-oz. size) of each. The chemicals listed or the equivalent shall be used. (See 6.6.)

3.3.1.1 OH number. The OH number for Pluracol TP-440 shall be between 383 and 413.

3.3.1.2 The stearic acid used in component I shall be of a reagent grade purity as supplied by the supplier. (See 6.6.)

3.3.1.3 The dibutyl tin diacetate used in component III shall be practical grade or higher in purity, as supplied by the supplier.

3.4 Mixing.

3.4.1 Component I (polyol). In 160 grams of Pluracol TP-440 shall be added 0.8 grams of stearic acid. The stearic acid shall be completely dissolved by gently heating the polyol between 55°C and 60°C (131° and 140°F) and stirring. Then add 2.4 grams of Dow Corning DC-113 and 48 grams of DuPont Freon F-11. The chemicals shall be stirred to insure

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complete mixing, making certain that any Freon that escapes is replaced. The polyol component shall then be stored in a suitable container (3.5.2) under an atmosphere of dry nitrogen until it is dispensed into the kit bottles. The stirring device shall not in any way react with the chemicals themselves.

3.4.2 Component II (isocyanate). In 190 grams of Mobay Mondur MR shall be added 21 grams of DuPont Freon F-11. The Freon shall be stirred into the Mondur MR until thoroughly mixed. Any Freon lost in the mixing process shall be replaced before storage. The isocyanate component shall be stored under a dry nitrogen atmosphere in a suitable container (3.5.2) until dispensed into the kit bottles. The stirring device shall not in any way react with the chemicals themselves.

3.4.3 Component III catalyst. The dibutyl tin diacetate catalyst shall be dispensed as received from the supplier into suitable containers (3.5.2) until dispensed into the kit bottles.

3.4.4 Scaling. The component I and II formulations may be scaled up to provide larger quantities provided that all properties of the foam chemical remain the same.

3.5 Mixing and storage devices.

3.5.1 Vessels. The vessels for mixing shall be selected to insure their complete inertness to the chemicals involved in the preparation of polyol and isocyanate components.

3.5.2 Containers. The containers used to store the polyol, isocyanate and catalyst until dispensed into the kit bottles shall not in any way react with the chemicals.

3.6 Bottles.

3.6.1 The mixed Components I and II chemicals shall be bottled separately. The bottles shall be of a pharmaceutical grade or equivalent in cleanliness, in 3 ounce size and of brown glass in an oval or round shape. Bottles shall be filled with 90 grams (± 1 gram). The air space above the chemicals shall be purged of air with dry nitrogen. In addition to the paper disc seal supplied in the plastic screw caps, the bottles shall be sealed with one mil thick FEP type A (chemical resistant grade) sheet of Teflon located between the cap and the top of the bottle. This sheet of Teflon shall be of such size as to completely cover the threads of the bottle as well as its opening when the cap is fitted. After the cap has been applied and tightened, excess Teflon sheet shall be trimmed to within 1/4 inch (± 1/8 inch) of the lower edge of the cap. The cap shall be as tightly affixed to the bottle top as possible without breaking or cracking the cap.

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3.6.2 Component III (catalyst). Twenty-five milliliters (± 5 ml) of catalyst shall be dispensed into 35-40 milliliter size bottles. The bottles shall be equipped with a rubber-bulb-type medicine dropper of polyethylene construction. The delivery orifice of the dropper shall have a mean diameter of approximately 1 millimeter. The rubber bulb shall be constructed of such material that will not contaminate the catalyst. It shall be capable of drawing up into the dropper at least the required quantity of catalyst for foaming one helmet liner (20 drops maximum). The caps of the bottles shall have a hole of such size as to admit the medicine dropper, giving a snug liquid-tight fit. The dropper shall be removable for shipment to enable utilization of the teflon seal. The bottle of catalyst shall be sealed in the same manner as the isocyanate and polyol. The seal shall be liquid tight. The air space above the catalyst shall also be purged with dry nitrogen prior to capping and sealing.

3.6.2.1 Alternate method of supplying catalyst. Twenty-five milliliters (± 5 ml) of catalyst shall be dispensed into 35-40 milliliter size flexible polyethylene bottles. These bottles shall be quipped with a Boston tip, or equivalent, whose delivery orifice shall have a mean diameter of approximately 1 millimeter. The Boston tip shall seat snugly in the bottle neck so that there will be no leakage of catalyst during use or inadvertent inversion during storage. The bottles shall be equipped with a plastic cap to seal off the Boston tip when not in use. This cap shall be of a screw-on type to provide a liquid-tight seal for storage.

3.7 Performance.

3.7.1 Quality of foam sample. When subjected to the tests specified in 4.6.1, the chemicals shall meet all requirements specified therein.

3.7.2 Weight. The weight of the polyol and isocyanate components shall be 90 grams, ± 1 gram, when tested as specified in 4.6.2.

3.7.3 Density - free foam. The density of the free foam shall be 2 pounds per cubic foot, ±0.25 pound per cubic foot, when tested as specified in 4.6.3.

3.7.4 Density - foamed custom fit liner. The density of the foamed custom fit liner shall be 3 pounds per cubic foot, ± 0.50 pound per cubic foot when tested as specified in 4.6.4.

3.7.5 Volume - catalyst. The volume of catalyst shall be 25 milliliters, ± 5 milliliters, when tested as specified in 4.6.5.1.

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3.8 Identification markings. Each component bottle shall be identified with the following information:

a. Specification number and date
b. Component number and name (1.2)
c. Batch number and date of manufacture
d. Weight and volume of material in bottle and volume of foam produced in the foaming process
e. Storage conditions: The bottle containers shall include the following information: "Store between 65°F and 75°F".

3.9 Toxicity. The components and resulting foam shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate departmental medical service who will act as an advisor to the procuring activity.

3.10 Odor. When mixing components there shall be no objectionable odor but the mixing process shall be performed in a well ventilated area.

3.11 Color. The polyol shall be clear and water white.

3.12 Workmanship. The workmanship shall be such as to insure a product which is uniform and in conformance with this specification. The material shall be homogeneous and free from contaminants and foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and tests required by this specification are classified as follows:

a. First article inspection
b. Quality conformance inspection.

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4.3 Inspection conditions. For each kit (see 6.5) of chemicals, the catalyst shall be added according to Table II.

Table II. Temperature vs. catalyst.

If temperature of component in °F is:	Number of drops to use:
60	20
65	18
70	16
75	14
80	12
85	11
90	10

Temperature shall be determined as follows:

a. Obtain an accurate temperature reading using a dial thermometer, such as a Weston Model 2265, or equivalent, immersed in one of the bottles.

b. At least 2 inches of the stem shall be immersed in the liquid.

c. Care shall be taken to insure the polyol and isocyanate are at the same temperature.

d. It is preferable that the chemicals be the same temperature as the room in which the foaming operation is taking place.

4.4 First article inspection. First article samples shall consist of 8 bottles of component I (polyol), 8 bottles of component II (isocyanate), and 1 bottle of catalyst. Five each first article samples submitted in accordance with 3.1 shall be subjected to the tests specified in 4.6.1, 4.6.2 and 4.6.3. Three each first article samples, submitted in accordance with 3.1, shall be subjected to the test specified in 4.6.4.

4.4.1 If any sample fails any first article test, all of the first articles submitted for test shall be rejected. Full particulars concerning the failure and action taken to correct the defect(s) shall be submitted to the procuring activity for concurrence and permission to repeat the first article test.

4.5 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.5.1 Lot. For the purpose of sampling, an inspection lot shall consist of material from one batch or run subjected to the same processing operation and conditions by the same manufacturer and presented for inspection at one time.

4.5.2 Sampling inspection. The inspection methods, the inspection levels, and the acceptable quality levels expressed in defects per 100 units for the sampling inspection shall be as listed in Table III. Random samples shall be drawn from each lot of end items.

Table III. Sampling inspection.

Characteristic	Inspection Level of MIL-STD-105	Acceptable Quality Level (Defects per 100 units)
Quality: Free foam (see 4.6.1)	S-2	2.5
Weight: Components I and II (see 4.6.2)	S-2	2.5
Density: Free foam (see 4.6.3)	S-2	2.5
Density: Foamed liner (see 4.6.4)	S-2	2.5
Examination for major defects (see 4.6.5)	S-2	4.0
Examination for minor defects (see 4.6.5)	S-2	6.5

4.5.3 Certification. The contractor shall furnish a certificate of compliance for the requirement specified in 3.3.1.1.

4.5.4 Resubmission of rejected lots. A rejected lot shall not be resubmitted for test and evaluation except with the approval of the contracting officer. The provisions listed under the paragraph entitled "Acceptance and Rejection" of MIL-STD-105 shall apply.

4.6 Inspection methods.

4.6.1 Quality of foam sample. The quality of the prepared formulation shall be tested by free-foaming a quantity equal to that used to prepare a liner. The foaming shall proceed as follows:

a. Weigh each bottle of polyol and isocyanate to the nearest 0.5 gram and record the weights.

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b. Pour the polyol component into an untreated 16 ounce paper cup.

c. Add the required amount of catalyst, as determined in 4.3.

d. Add the isocyanate component.

e. The mixture shall be vigorously stirred with a tongue depressor (approximately 3/4 inch wide by 6 inches long) until the liquid turns a dark brown to black in color. This color change shall occur within 15 to 20 seconds after the beginning of the stirring action.

f. Stirring shall then cease after the color change and the contents observed for the beginning of foam reaction which shall occur 8 to 15 seconds after cessation of stirring.

g. Completion of the foam reaction shall occur within approximately 20 seconds to 1 minute after initial foam reaction.

h. The upper surface of the mass of foam shall be dry to the touch within 2 minutes after completion of foam reaction and cured to the point of resisting finger impression within 5 to 10 minutes after completion of foam reaction.

i. Shrinkage of the cured foam shall be minimal so as to have no effect on the fitting of a foamed liner within an Air Force flight helmet.

j. A section of the resultant foam shall be tested for density as specified in 4.6.3.

4.6.2 Weight. The empty weight of the polyol and isocyanate bottles, including their caps and Teflon sheet material, shall be substracted from the weights determined in 4.6.1, to determine conformance to 3.7.2.

4.6.3 Density - free foam. The density of the free foam shall be determined as follows:

a. A sample section of the dry foam shall be weighed to the nearest 0.01 gram.

b. An empty, clean beaker, large enough to contain the sample shall be placed in a flat pan of at least the same volume as the beaker.

c. The beaker shall be filled with tap water to the overflow point. Care shall be taken to insure against overflow.

d. The sample foam shall be immersed completely in the water and the displaced water caught in the pan.

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e. The displaced water shall be poured into a graduated cylinder and its volume recorded to the nearest cubic centimeter.

f. The density of the foam in pounds per cubic foot shall be determined in accordance with the following formula:

$$\text{Density in pounds per cubic foot} = \frac{\text{sample weight in grams}}{\text{volume of displaced water (in cubic centimeters)}} \times 62.45 \ \frac{cm^3 lbs}{gm.ft^3}$$

4.6.4 Density - foamed custom fit liner. Custom fit liners shall be poured utilizing a suitable head form and the medium size KMU-439/P liner mold, FSN 8475-00-166-0128. Density shall be determined as noted in 4.6.3. Procedures for foaming the helmet liner are stated in the Appendix.

4.6.5 Examination for major and minor defects. The bottles and chemicals shall be examined for the major and minor defects listed in Table IV.

Table IV. Classification of major and minor defects.

No.	Defects	Major	Minor
1	Bottles (Components I and II) containing less than 89 grams or more than 91 grams.	*	
2	Bottles (Component III) containing less than 20 milliliters or more than 30 milliliters.	*	
3	Teflon sheet not visible under bottle cap.	*	
4	Bottle caps cracked or broken.	*	
5	Bottles showing evidence of leakage.	*	
6	Chemicals showing evidence of separation into layers, thickening, hardening or precipitation.	*	
7	Any marking missing, incorrect, illegible or incomplete.		*

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4.6.5.1 Weight of the polyol and isocyanate shall be determined in accordance with 4.6.2. The volume of catalyst shall be determined by pouring the bottled contents into a graduated cylinder and recording the volume to the nearest milliliter. Bottles utilized for these tests shall be over and above those to be delivered on contract.

4.6.6 Packaging inspection. An examination shall be made to determine that the preservation-packaging, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. Defects of colsure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Material	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Content	Number of bottles per shipping container is more or less than required.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. The components I, II and III containers and the quantities contained shall be as specified in this specification.

5.1.1.1.1 The glass bottles will be closed with screw caps conforming to the requirements of this specification. After sealing the containers, caps shall have such torque applied as to seal the bottles against leakage and loosening during handling, shipment and storage.

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5.1.2 Level C.

5.1.2.1 Unit packaging. The components I, II and III shall be packaged in unit capacity containers in accordance with the suppliers commercial practice.

5.2 Packing. Level of packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Bottles. Components I, II, and III shall be packed in accordance with PPP-G-460.

5.2.2 Level B.

5.2.2.1 Bottles. Components I, II and III shall be packed in accordance with PPP-G-460.

5.2.3 Level C.

5.2.3.1 Components I, II and III packaged as specified in 5.1.2.1 shall be packed in exterior containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with Uniform Freight Classification Rules, or regulations of the National Motor Freight Classification, as applicable.

5.3 Marking of shipments. Interior and exterior shipping containers shall be marked in accordance with this specification (3.8) and in accordance with MIL-STD-129. The following precautionary marking shall appear on two opposite sides of each exterior container whenever practicable, depending on the size of the carton.

STORE IN A COOL PLACE
GLASS - DO NOT THROW OR DROP

6. NOTES

6.1 Intended use. The plastic material of cellular polyurethane, foam-in-place is intended to be used in helmets as a liner.

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6.2 Ordering data. Procurement documents should specify the following

a. Title, number and date of this specification.

b. Component designations.

c. Quantity desired.

d. Level of packing required.

6.3 Unit of purchase. The unit of purchase shall be a carton containing 50 kits (see 6.5). Separate purchase of components may be procured, if specified.

6.4 Government reference. The Government activity responsible for the foam-in-place preparation formula is the AFWAL/MLSA, Wright-Patterson Air Force Base, Ohio 45433.

6.5 Kit. A kit or set shall consist of one bottle of polyol and one bottle of isocyanate. A bottle of catalyst shall be supplied with each unit of purchase.

6.6 Chemicals. The recommended source for chemicals used to prepare the foam components have been found to be satisfactory, and may be obtained from the listed suppliers, or the equivalent may be used.

Component	Material	Supplier
I (Polyol)	Pluracol TP-440	Wyandotte Chemical Co.
	Silicon DC-113	Dow Corning Corp.
	Stearic acid (reagent grade) Stock No. A-293	Fisher Scientific Co.
	Freon F-11	E. I. DuPont de Nemours
II (Isocyanate)	Mondur MR	Mobay Chemical Co.
	Freon F-11	E. I. DuPont de Nemours
III (Catalyst)	Dibutyl tin diacetate (6108-P - practical)	Fisher Scientific Co.

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6.6.1 Bottles. Government activities shall procure bottles identified as FSN 6530-782-6431, or the equivalent. Nongovernmental activities shall procure bottles of excellent quality from commercial sources.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

(Project No. 8030-F032)

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APPENDIX

PROCEDURES FOR USE OF FORM-FIT HELMET LINER MOLD AND FOAMING PROCESS

10. SCOPE

10.1 Scope. The procedure for use of the mold are outlined in step-by-step procedures.

20. PART I

20.1 The system. There are six basic parts to the helmet mold system and they are as follows: Teflon coated mold shell, adjusting ring, spacer cap, seal ring, ejector tool assembly and the mold case.

20.1.1 The rubber spacer cap and seal ring are to be coated with Dow Corning Dispersion Coating No. 92-009 or the equivalent, after approximately 6 to 10 foamings.

20.1.2 The dispersion agent will prevent the polyurethane foam from sticking to the rubber parts and causing deterioration. The procedure for repeating this process on the rubber items will be outlined later.

20.2 Fitting the mold.

Step a. Place the adjusting ring over a suitable head form with the strap adjusting screws (labeled "ADJ" and "LOCK") to the rear. It should be centered on his head so the ring covers his eyes. The height of the ring is controlled by the strap. To adjust the strap, first loosen slightly the screw marked "LOCK" adjust with the screw marked "ADJ", then finger tip tighten "LOCK" screw. Now place the Teflon coated mold shell on the adjusting ring. With a ruler, measure the distance from the of the subject's head to the top of the Teflon coated shell. It should be approximately 1-1/4 inches. Adjust the strap to achieve the desired distance if necessary. Remove the mold shell and put the dispersion release agent-coated spacer cap on the head form OVER the straps. The front of the spacer cap is where the seams on the inside come together. Pull it down tightly and fasten under the chin with the velcro. Smooth out all air under the spacer cap by smoothing the cap with your hands from the center of the subject's head out and down in all directions. Check to see that the adjusting ring is still centered.

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Step b. Tighten the four adjusting screws in the adjusting ring so that the centralizing bars exert equal and gentle pressure against the spacer cap and the head form. The ring should be approximately equal distance from the spacer cap all around. This step insures that the ring will not move during the foaming process.

Step c. Next, place the dispersion release agent coated seal ring over the spacer cap, pulling it evenly all the way down to the adjusting ring being careful not to disturb the centering of the adjusting ring. The seal ring has no top or bottom and either side can have the dispersion coating applied. The narrower end of the seal ring is the front. Recheck the centering and spacer cap fit and see that the seal ring is laying approximately flat on the surface of the adjusting ring.

Step d. The Teflon coated mold shell has a line of dots on the inside surface. This should be traced with the red grease pencil, this is the trim line after foaming and the heat of the foaming process will transfer the line to the foam liner.

Step e. Put the mold shell on the seal ring and locate the shell with the three fastener location bolts and evenly tighten them down finger tight.

30. PART II

30.1 Foaming process. The room temperature should be maintained at 75°F (± 5°F). The mold and chemicals should be at room temperature also. The protection coat supplied with the system is to be worn by the subject during the process. The foam and chemicals will ruin clothes since it cannot be removed. A well ventilated area is necessary because of the isocyanate gas produced by the chemical reaction.

Step a. Pour the contents of the bottle of polyol into a paper cup. Add catalyst in accordance with 4.3, Table II, from the eye dropper. The drops should be full drops with no air to insure proper volume of catalyst. Avoid getting the catalyst into the bulb of the dropper as it will deteriorate it. Flush the bulb with water and dry after making sure dropper and bulb are dry before reusing.

Step b. Add the contents of a bottle of isocyanate (the dark colored liquid) and stir immediately with the wooden tongue depressor. Continue stirring rapidly until the mixture changes to a very dark

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brown almost black color. Rapidly discard the stirring stick and hold the cup tilted above the opening an the mold shell. When the mixture starts to froth on the surface, and it has risen about 1/4 inch above the level of the liquid, pour it rapidly, but smoothly into the mold opening, starting at the front and pouring toward the back and working to the front again using all the liquid.

Step c. Quickly discard the mixing cup and rotate the head form and mold slightly to distribute the liquid evenly.

Step d. The foam should rise rapidly (about 10 seconds) filling the entire mold. It should expel a small amount of foam through each vent hole and rise 1 inch to 3 inches out of the large pouring opening.

Step e. After about 5 minutes, when the foam is not sticky to the touch, you can start to pinch and tear off the small parts of foam that has expelled through the vent holes. Cut the excess foam that has expelled through the top opening off to approximately 1/4 inch above the Teflon coated mold shell being careful not to cut the Teflon (This can be done best sometimes by cutting to within 1/2 inch of the mold shell in the first cut, then cutting to within approximately 1/4 inch of the mold shell.)

40. PART III

40.1 Removing the mold and liner. Remove the mold from the head form by first loosening the velcro chin strap, and backing off the ring adjusting bar screws completely. Then carefully and slowly lift the whole mold straight up and off the head and set the mold on the ball supports for approximately 5 minutes. (Total time to this point is approximately 13 to 15 minutes from pouring.)

40.2 Carefully remove the spacer cap from the mold. With the mold setting on the ball support feet, loosen the three location fastener bolts until they are swung out of the way to remove the mold shell. Peel off the seal ring.

40.3 By holding the mold shell with the foam still in it (between your hands as you would hold a basketball - with the open side of the shell toward you), squeeze the shell causing it to flex, rotate the shell 90° and repeat. This will break the foam loose from the shell. Rotating the shell and flexing it a few times will help with removal of the foam liner. Excess force is not required and could damage the mold shell. A distinct cracking can be heard indicating release of the foam from the shell. Place the shell in the mold case with the ejector connection toward the rear of the case and the three bosses

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resting in the contents. Insert the ejector tool in the connection and line the ejector pad central over the form pour opening.

40.4 Press down on the ejector handle and the liner will eject into the case.

40.5 Clean the mold and all components of any foam or residue by wiping or peeling it away being careful not to scratch the Teflon covered shell with anything sharp.

40.6 Questions on the process or mold should be referred to: AFWAL/MLSA, Wright-Patterson Air Force Base, Ohio 45433.

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